

HDOC - 36

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Question

Write a menu-driven C program to perform the following operations using separate functions:

1. Count characters in a string without using built-in length functions.
2. Print each character of a string on a new line.
3. Count vowels and consonants in a string.
4. Exit.

Program

```
#include <stdio.h>
#include <ctype.h>

int countCharacters(char str[]) {
    int i = 0;
    while (str[i] != '\0')
        i++;
    return i;
}

void printCharacters(char str[]) {
    int i = 0;
    while (str[i] != '\0') {
        printf("%c\n", str[i]);
        i++;
    }
}

void countVowelsConsonants(char str[]) {
    int i = 0, vowels = 0, consonants = 0;
    char ch;

    while (str[i] != '\0') {
        ch = tolower(str[i]);

        if (ch >= 'a' && ch <= 'z') {
            if (ch=='a' || ch=='e' || ch=='i' ||
                ch=='o' || ch=='u')
                vowels++;
            else
                consonants++;
        }
        i++;
    }

    printf("Number of vowels = %d\n", vowels);
    printf("Number of consonants = %d\n", consonants);
}

int main() {
    char str[100];
    int choice;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    /* Remove newline added by fgets */
    int i = 0;
    while (str[i] != '\0') {
        if (str[i] == '\n') {
            str[i] = '\0';
            break;
        }
        i++;
    }

    do {
        printf("\n--- MENU ---\n");
        printf("1. Count characters\n");
        printf("2. Print each character on a new line\n");
        printf("3. Count vowels and consonants\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
```

```

scanf("%d", &choice);

switch (choice) {
case 1:
    printf("Number of characters = %d\n",
           countCharacters(str));
    break;

case 2:
    printf("Characters are:\n");
    printCharacters(str);
    break;

case 3:
    countVowelsConsonants(str);
    break;

case 4:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice\n");
}
} while (choice != 4);

return 0;
}

```

Sample Output

Enter a string: Hello

--- MENU ---

1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 1

Number of characters = 5

Enter your choice: 2

Characters are:

H
e
l
l
o

Enter your choice: 3

Number of vowels = 2

Number of consonants = 3

Enter your choice: 4

Exiting program...

Result: The program successfully counts the characters without using a built-in length function, prints each character on a new line, and counts vowels and consonants using separate functions.